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FACULTY OF ELECTRICAL AND ELECTRONIC S ENGINEERING TECHNOLOGY

BVI 3114 TECHNOLOGY SYSTEM OPTIMAZATION II
Problem-Based Learning

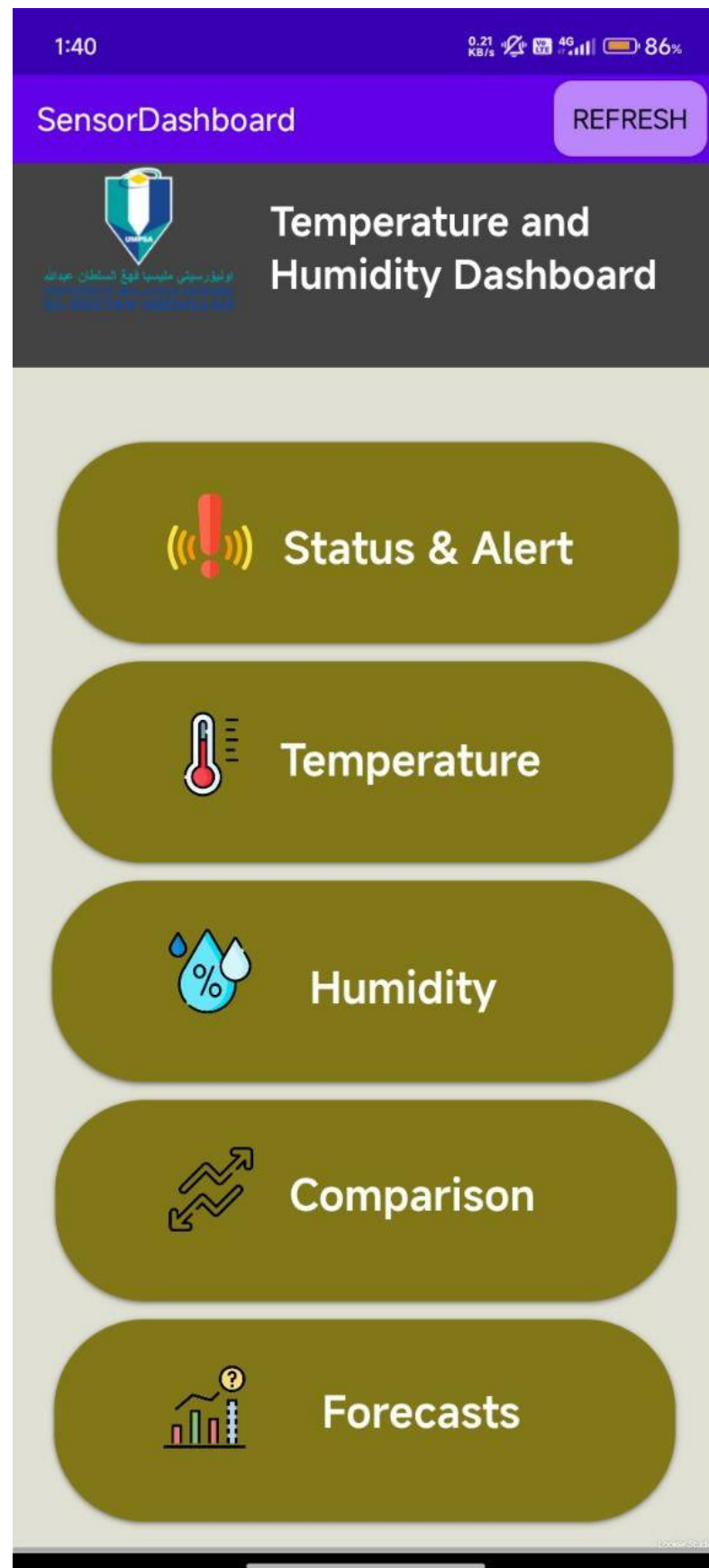
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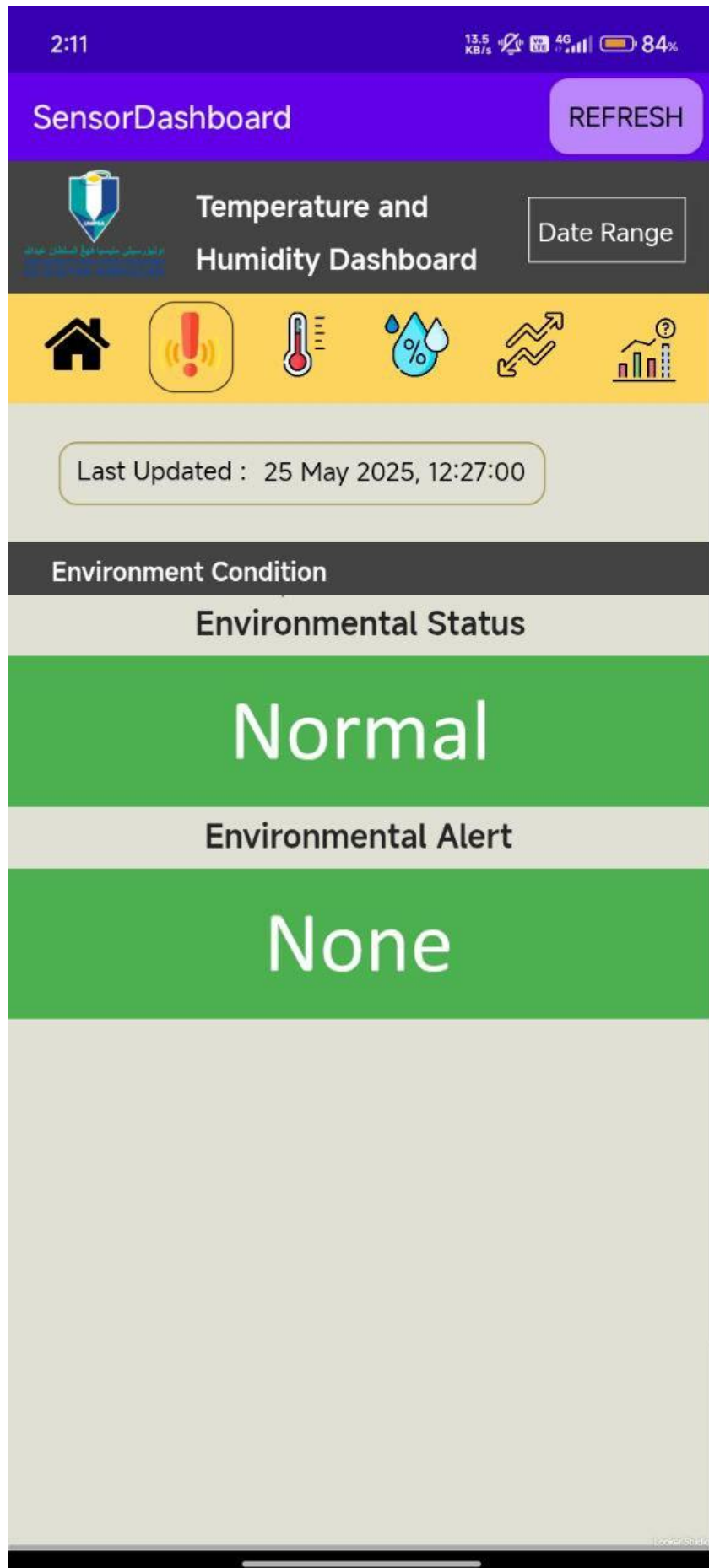
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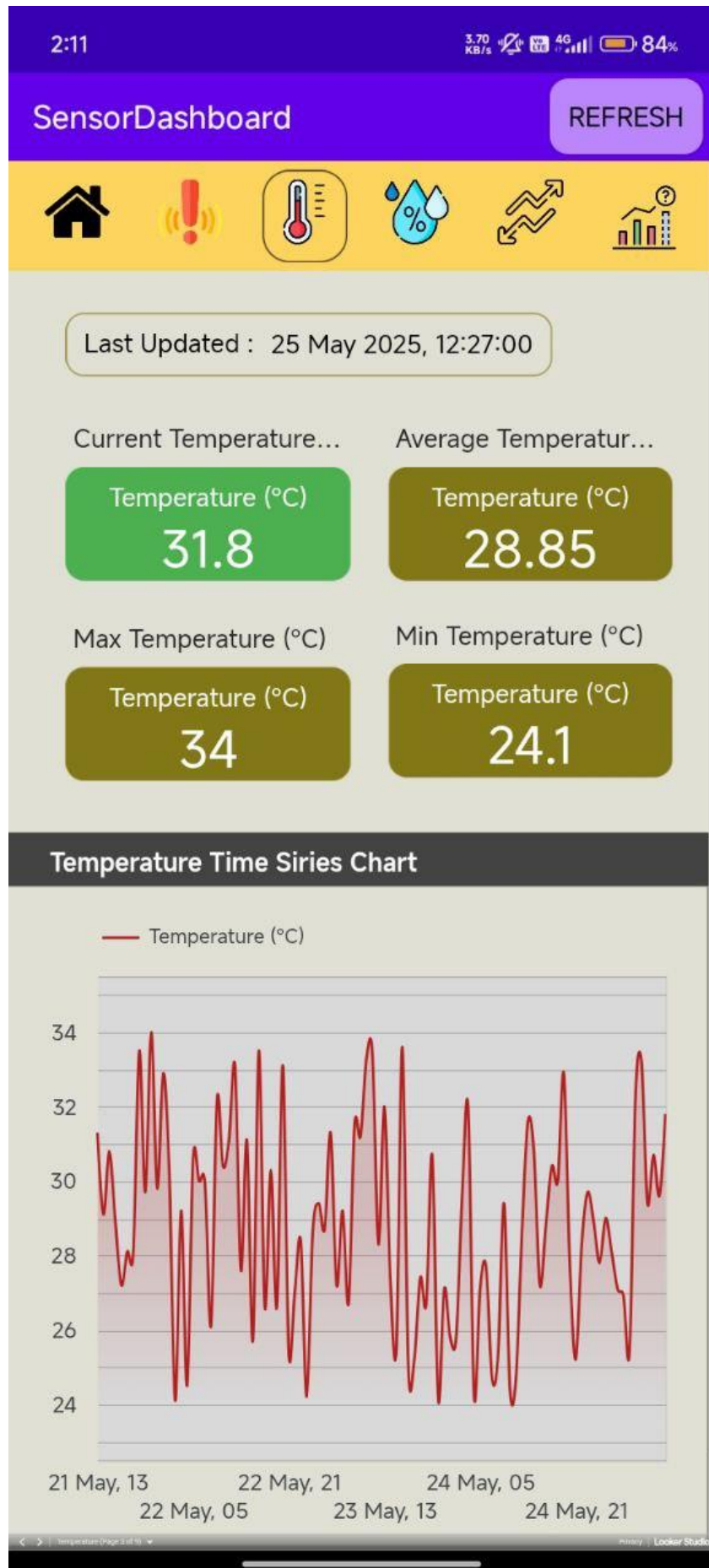
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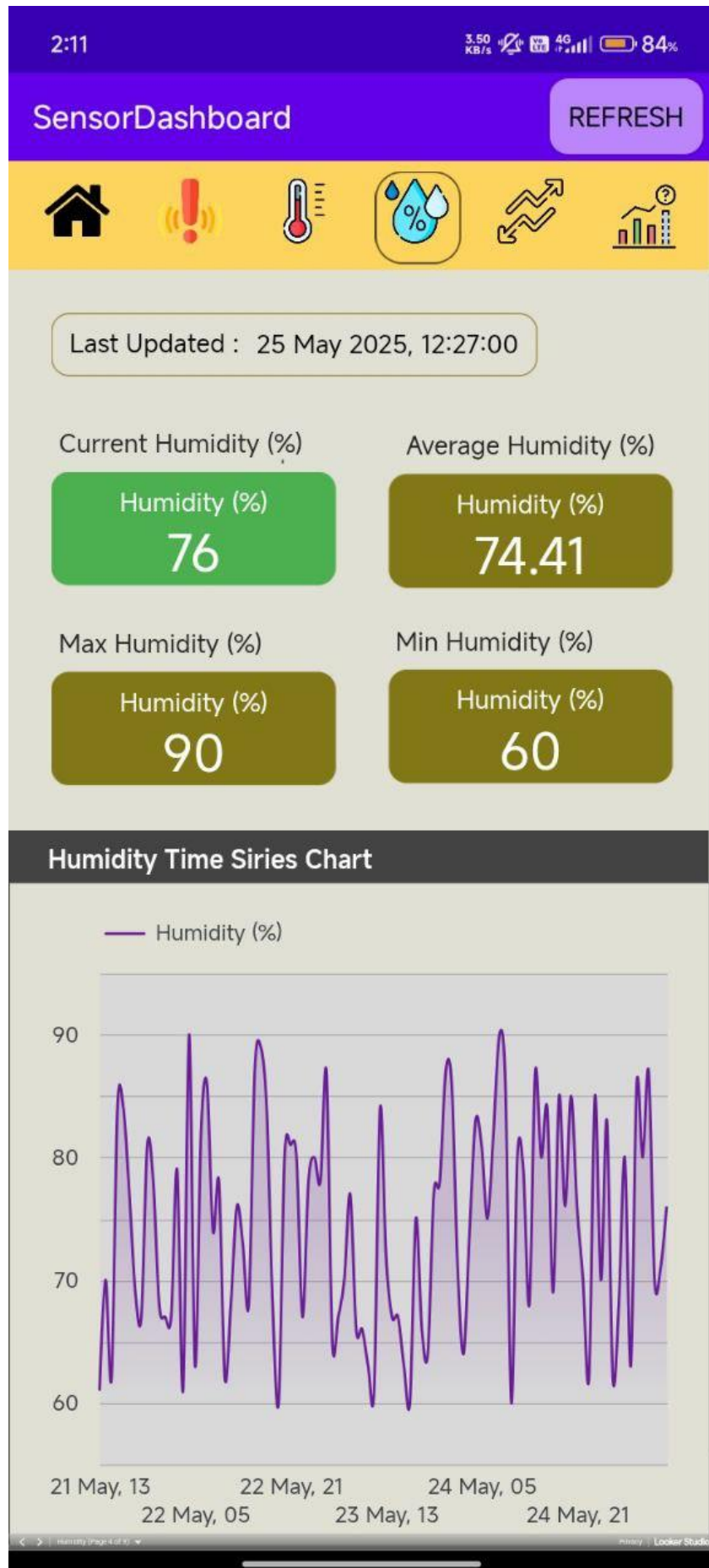
1. GitHub Repository Link
<https://github.com/Symsl23/PBL-Forecasting-TSO2.git>
2. Google Sheet Forecasting Data Link
<https://docs.google.com/spreadsheets/d/1Tv1dCwnZ2F99xOxNqF-gMaQwryHTmjH3GWY5vIGP7IE/edit?usp=sharing>
3. Looker Studio Dashboard Link
<https://lookerstudio.google.com/reporting/539c465f-5813-4fad-b333-68e2e341b934>

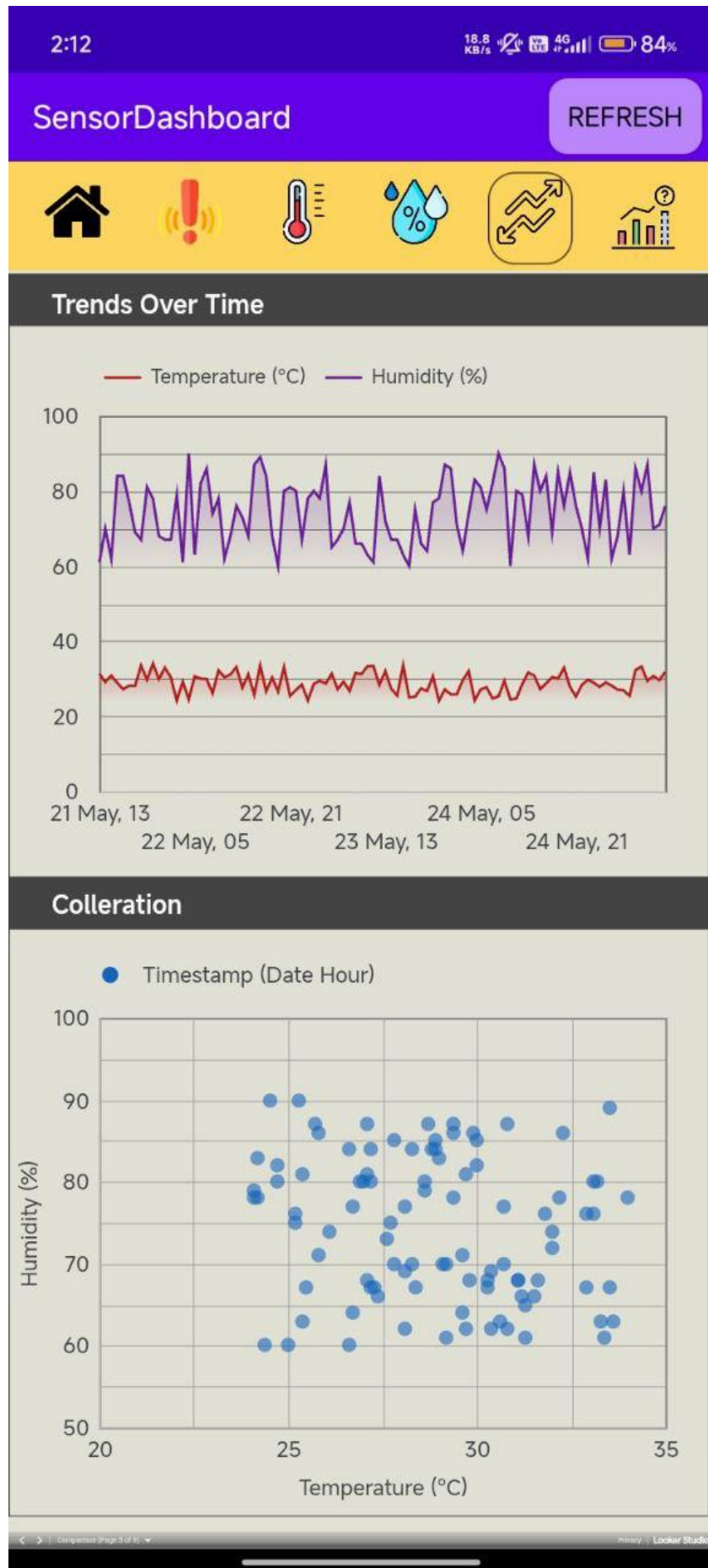
4. Complete Mobile App Screenshot

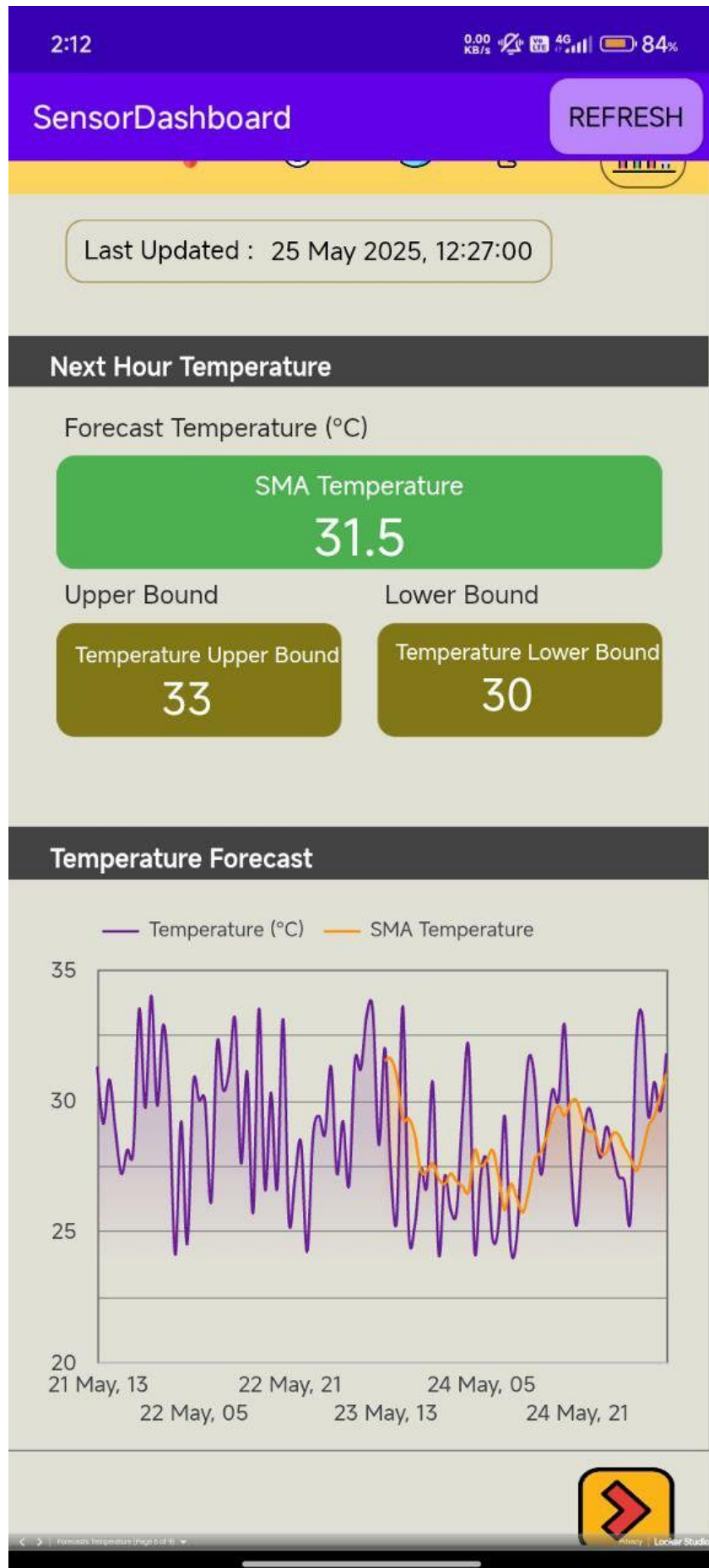


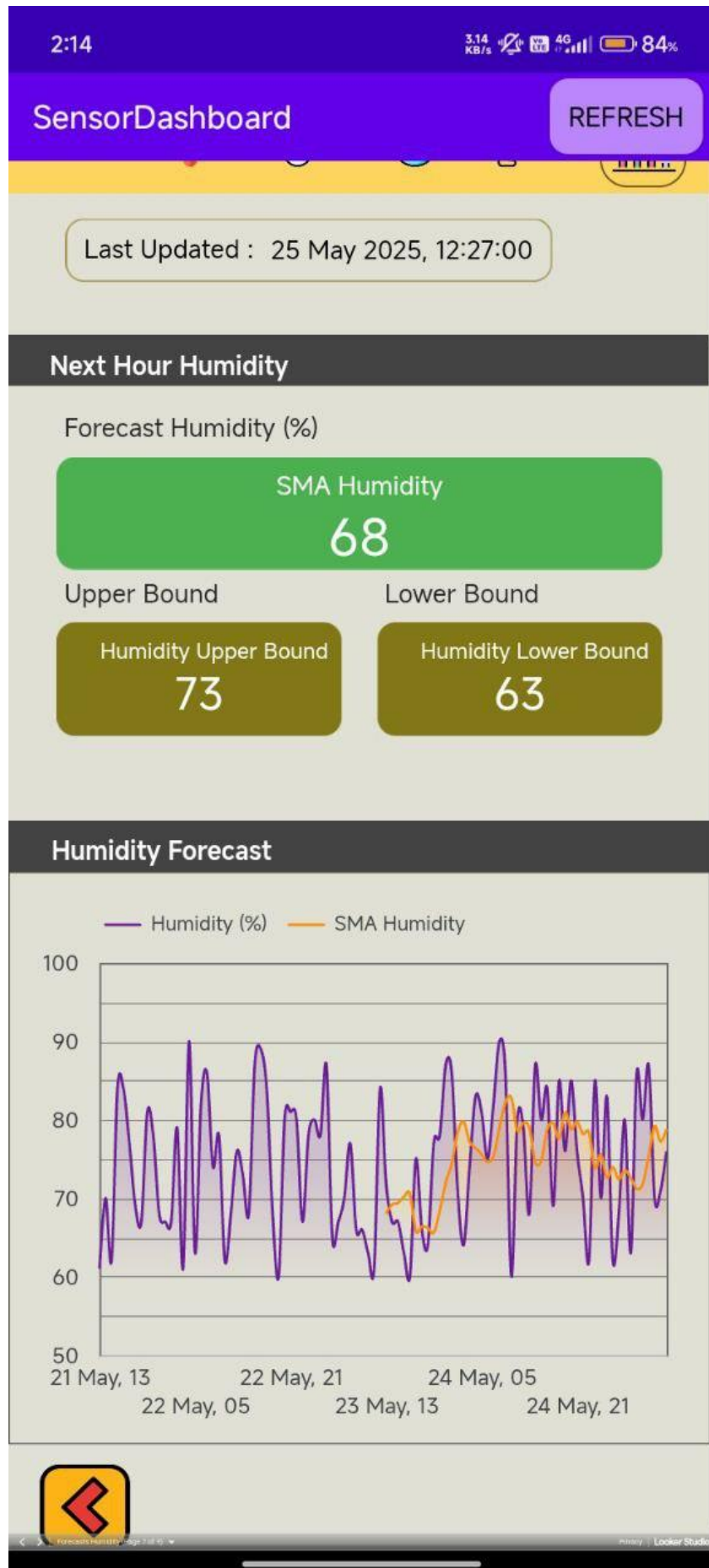


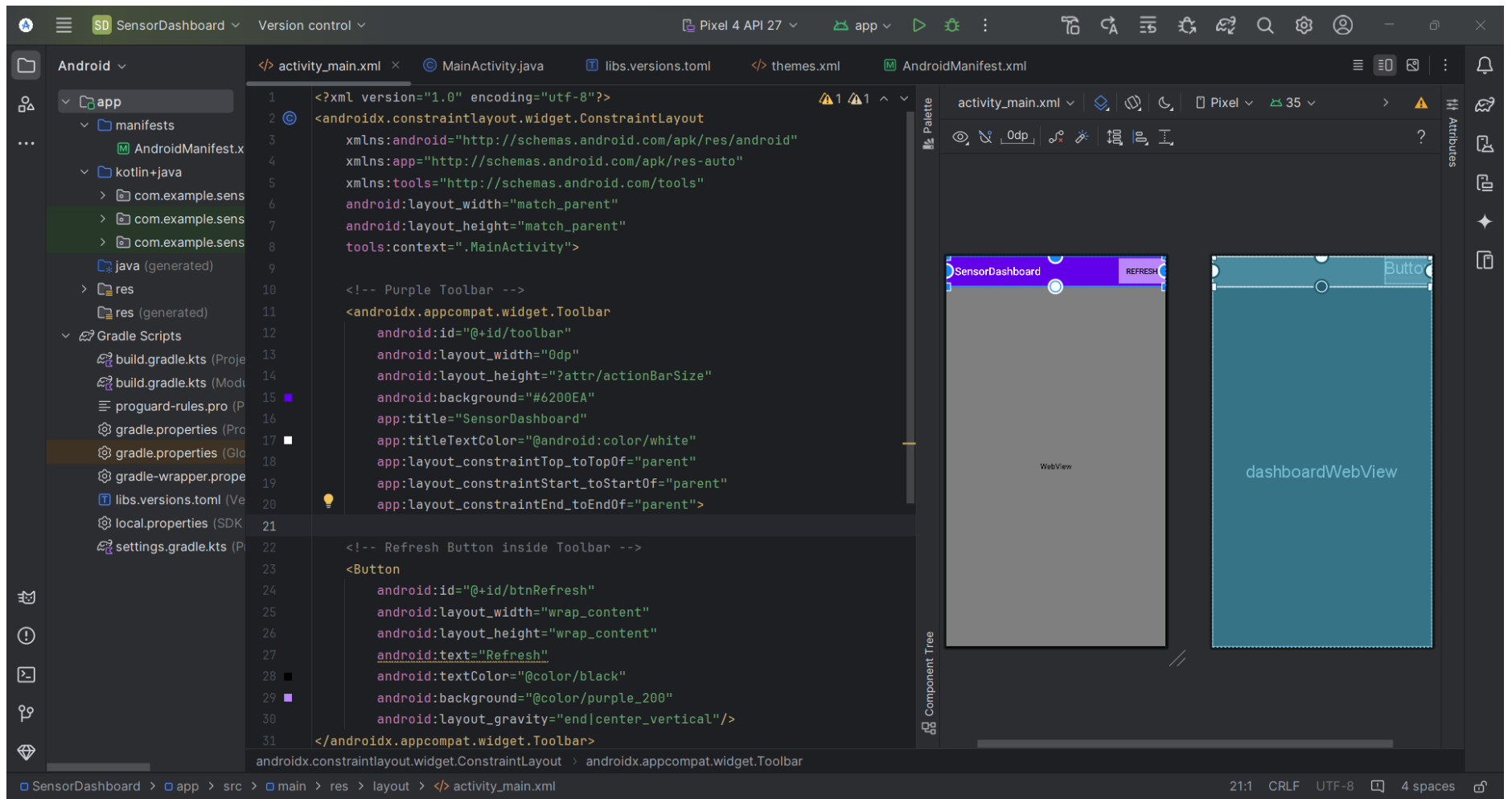










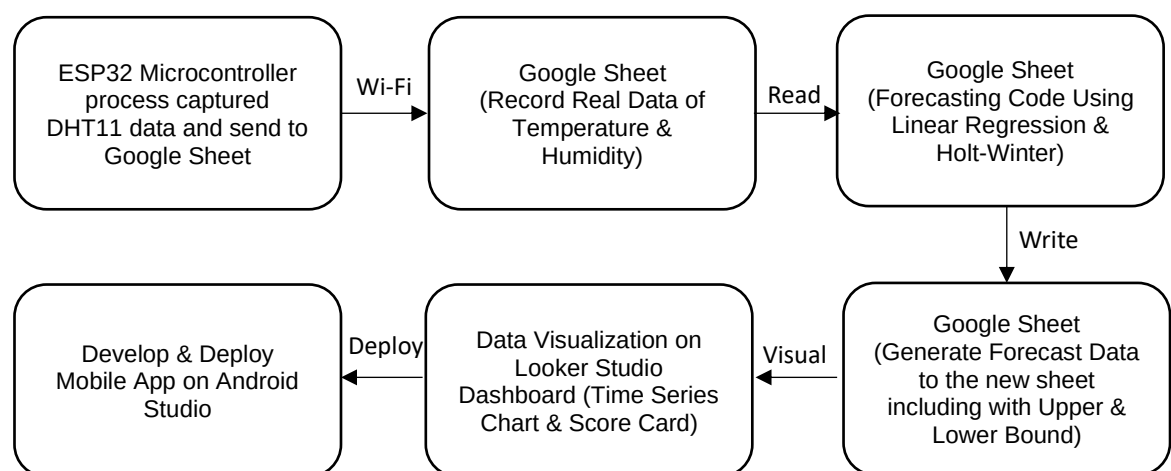


5. Documentation

5.1 Executive Summary

This project is an integrated IoT-based solution that capture sensor data from an ESP32 microcontroller and applies forecasting algorithms using Google Apps Script to predict future values. The objective is to enhance an existing sensor monitoring system with predictive analytics to better support decision-making. The system visualizes both historical and forecasted data in Google Looker Studio, offering insights into trends and future conditions. This project highlights effective integration of hardware, data-based scripting, data visualization platforms and mobile app deployment.

5.2 System Architecture Diagram



5.3 Forecasting Algorithm Explanation

5.3.1 Simple Moving Average (SMA)

Definition

The Simple Moving Average is a basic forecasting method used to smooth out short-term fluctuations and highlight longer-term trends in data. It calculates the average value of the most recent readings from a sensor.

App Script Implementation

```
function forecastTemperatureHumiditySMA() {  
  const sheet = SpreadsheetApp.getActiveSpreadsheet().getActiveSheet();  
  const tempData = sheet.getRange("B2:B97").getValues().flat(); // Temperature data  
  const humData = sheet.getRange("C2:C97").getValues().flat(); // Humidity data  
  const windowSize = 5;  
  
  for (let i = 48; i < 96; i++) { // Day 3 and 4: rows 49 to 96  
    if (i < windowSize) continue;  
  
    // Temperature calculations
```

```

const tempWindow = tempData.slice(i - windowSize, i);
const tempAvg = tempWindow.reduce((a, b) => a + b, 0) / windowSize;
const tempUpper = tempAvg + 1.5;
const tempLower = tempAvg - 1.5;

// Humidity calculations
const humWindow = humData.slice(i - windowSize, i);
const humAvg = humWindow.reduce((a, b) => a + b, 0) / windowSize;
const humUpper = humAvg + 5;
const humLower = humAvg - 5;

// Write to columns D to I
const row = i + 2;
sheet.getRange(row, 4).setValue(tempAvg.toFixed(1)); // Column D
sheet.getRange(row, 5).setValue(tempUpper.toFixed(1)); // Column E
sheet.getRange(row, 6).setValue(tempLower.toFixed(1)); // Column F
sheet.getRange(row, 7).setValue(humAvg.toFixed(1)); // Column G
sheet.getRange(row, 8).setValue(humUpper.toFixed(1)); // Column H
sheet.getRange(row, 9).setValue(humLower.toFixed(1)); // Column I
}
}

```

Benefits

- **Simple Forecasting Technique**
SMA predict the next value based on recent data trends and its good for short-term forecasting
- **Reducing noise in sensor data**
SMA help to smooth the minor fluctuation and sensor noise of the sensor data and giving a more stable value
- **Easy to Implement**
SMA only need a basic math (average calculation) to predict the forecast value of the data and requires a minimal resource like ESP32 and Arduino Uno

Limitations

- **Poor for Long-Term Forecasting**
It works well only for short term data prediction and not suitable for long-term prediction trend or irregular patterns
- **Cannot handle Missing or Irregular Data**
Gaps or inconsistent reading can affect the average value

5.4 Implementation Challenges and Solutions

- i) The historical data and forecasted data are not shown properly in the chart because of Date/Time dimension not recognized as a proper date/time field.
Solution :
Recognized the Date/Time field format to the Date Hour Minute, so that the chart will show all data from the sheet.
- ii) Inability to visualize Linear Regression and Holt-Winters forecasts on the same chart is due to them being in separate sheets.
Solution :
Manually copy & paste the both forecasted data in one sheet then add the data source to the Looker Studio again.
- iii) Linear Regression Algorithm are not suitable for the DHT11 because the forecasted value not shown a pattern only the straight line
Solution :
Change the algorithm into Simple Moving Average (SMA). This algorithm is more suitable because it using a basic math of average calculation and can reduce noise for the DHT11 data

5.5 Results and Future Improvements

The result of this system is shown on the Complete Mobile App Screenshot above, below is the future improvement of our system.

- i) Model Comparison and Accuracy Metrics
To ensure the forecasting are reliable and justifiable, implement model performance metric are required like MAE (Mean Absolute Error) to measure average of error, RMSE (Root Mean Squared Error) to give penalties to the bigger error and MAPE (Mean Absolute Percentage Error) is useful for percentage-based error analysis.
- ii) Add Anomaly Detection
The system will get benefit of detecting an abnormal sensor and forecast reading by compare historical data with forecasted value and trigger alert or indicator if the data exceeds a threshold.
- iii) Use More Advanced Forecasting Algorithms
The current algorithm uses a Linear Regression and Holt-Winter, more advanced algorithm can offer better accuracy and adaptability to complex pattern. The potential algorithm to be improved is using an ARIMA (Auto Regressive Integrated Moving Average) that suited for stationery time series data, capturing both trend and auto colleration.